MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE

EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCBCH100 - BIOCHEMISTRY-THE SCIENCE OF LIFE

(2024 ADMISION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and

Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome

attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. Which of the following is NOT a feature of eukaryotic cells?
   Ans-Nucleoid region
2. Identify the monosaccharide among these
   Ans- Option c) Glucose
3. Atom most commonly involved in hydrogen bonding
   Ans- Option a) Oxygen
4. Which of the following hetero polysaccharides is known for its role in the skin and joint lubrication?
   Ans-b) Hyaluronic acid
5. What is the function of ribosomal RNA (rRNA)?
   Ans- Option c) forms the structural and catalytic components of ribosomes.

[5x1=5]

Part B

Fill in the Blanks

Answer **All** Questions

Each Question carries **2** marks

6. The Kw value for water at 25°C is $\underline{1 \times 10^{-14} M}$but at higher temperatures this value<u>increases</u>

7. As the concentration of hydroxide ions increases, the pOH value decreases.
8. The formation of ring structure in monosaccharides creates a new center of asymmetry, this results in 2 isomeric forms called anomers
9. Which is the storage form of glucose in animals- glycogen
10. The structure of nucleotides allows them to store/carry/disseminateinformation in genetic material.

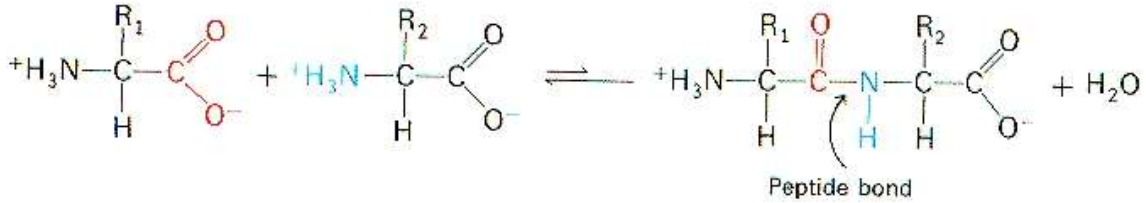
[5x2=10]

Part C

Short Answer Questions
Answer any **5** Questions
Each Question carries **3** marks

11. Explain monounsaturated fattyacid and give one example with structure
    Definition- 1 mark
    Example -1 mark
    Structure- 1 mark
12. Write a note on peptide bond.

    Peptide bonds are covalent bonds that link the carboxyl group of one amino acid to the amino group of the next amino acid in the sequence. This results in a long chain of amino acids with a repeating N-C-C-N-C-C backbone.



13. Justify that structural features allow DNA to carry hereditary information.

    Genetic information is encoded in the precise sequence of bases in the DNA strands.

    Segments of DNA that carry genetic information are called genes.

    Explanation of chemical structure of DNA-**3 marks**

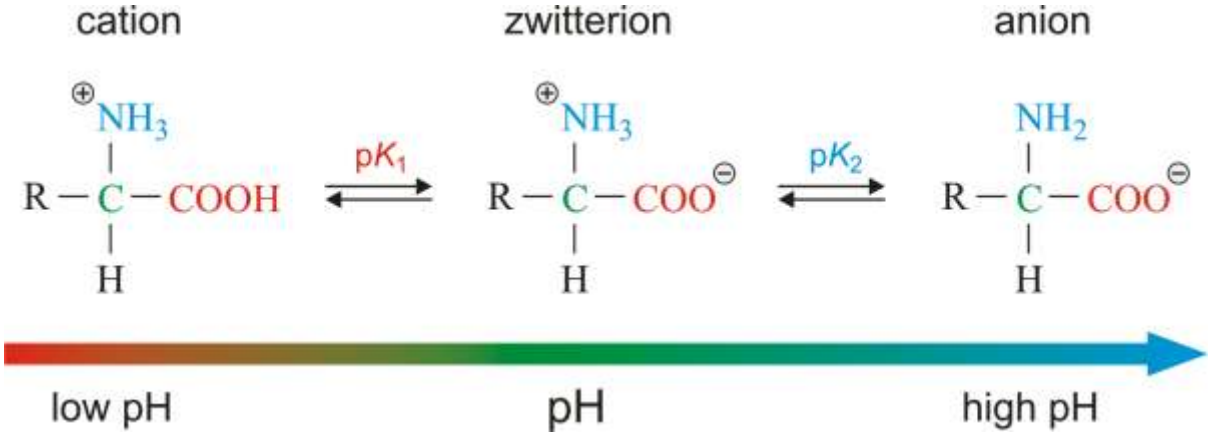
14. In what ways does biochemistry influence nutrition?
    Anything related to nutrition is written- 3marks
15. Differentiate between monosaccharides, disaccharides and polysaccharides, with examples of each.
    1 mark each
16. What is Biochemistry?
    Suitable explanation of Biochemistry- 3 marks.
17. Summarize the different classes of lipids.
    Classification – 3 marks

[5x3=15]

Part D

Short Essay Type Questions
Answer any **4** Questions
Each Question carries **5** marks

18. Explain the significance of the pKa value in the Henderson-Hasselbalch equation. How does the pKa value affect the pH of a buffer solution.
    (For attempt give full marks)
19. What are the main types of RNA and their functions.
    Name of 3 types of RNA -1/2 mark each
    Functions of 3 types of RNA- 1 mark each
    If full marks obtained for the above (4.5 marks) give additional ½ marks
20. What are aminoacids? Mention the one letter and three letter codes of twenty standard aminoacids.
    one letter and three letter codes of 4 amino acids- 1 mark, 2o amino acids grouped into 5 groups of 4 each (1 x 5 =5)
21. Explain the concept of zwitterion and its main properties.
    Explanation -2 marks
    Illustration- 3 marks



22. Differentiate the structure of triacyl glycerol. How does its composition relate to its role in energy storage.
    Definition and structure-2.5 marks
    (composition related to role in energy storage) If the question is attempted give 2.5 marks.

23. Explain the structure and complementary base pairing of DNA.
    1 mark each for a point

[4x5=20]